Statistical Analysis Plan (SAP)

TITLE:

STUDY DRUG: CMP-135

INDICATION: Ovarian Cancer

IND:

SPONSOR: Company, Inc

DATE FINAL:

DATE AMENDED:

OBJECTIVES

Primary Objective

The primary objective for this study is as follows:

- To estimate the clinical benefit of the addition of CMP-135 as maintenance therapy for patients with ovarian cancer in a second or third complete remission, as measured by investigator-determined PFS by radiographic assessment

Secondary Objectives

Secondary objectives for this study are as follows:

- To make a preliminary assessment of the safety and tolerability of CMP-135 in this population
- To make a preliminary assessment of the efficacy of CMP-135 as measured by overall survival

STUDY DESIGN

The proposed study is a Phase II, randomized, placebo-controlled, double-blind, multicenter clinical trial of CMP-135 in patients with ovarian cancer in a second or third complete remission. Patients will be randomized in a 1:1 ratio to either CMP-135 or placebo. Randomization will be stratified based on whether their cancer is in a second or third complete remission. Patients will receive 150 mg of CMP-135 or placebo daily by mouth. Treatment will continue until evidence of radiographic progression, until the patient experiences intolerable toxicities most probably attributable to CMP-135, or until the patient withdraws from the study. Day 1 of the study will be defined as the first day the patient receives CMP-135 or placebo. Patients who discontinue study treatment (i.e., due to documented disease progression) will be followed for survival approximately every 3 months until death, loss to follow-up, or study termination by Company.

OUTCOME MEASURES

Primary Efficacy Outcome Measure

The primary outcome measure is PFS, defined as the time from randomization to the first occurrence of radiographic progression or death from any cause, determined by the investigator. Radiographic progression is defined as the appearance of a new lesion on a computed tomography (CT) scan of the chest/abdomen/pelvis consistent with recurrent disease, including new onset or worsening ascites. New ambiguous lesions may be biopsied to determine progression. The patient should remain on blinded study treatment (CMP-135 or placebo) until radiographic progression is confirmed. In the case of clinical deterioration, such as small bowel obstruction, a CT scan should be performed to determine whether radiographic progression has occurred. If radiographic progression cannot be documented objectively, the investigator must consult the Medical Monitor before removing a patient from the study. Increasing levels of CA-125 in the absence of radiographic progression has not been validated as an endpoint in this setting and should not be interpreted as progression or as a criterion for discontinuation of study treatment.

For patients who discontinue study treatment prior to documented radiographic progression, every effort should be made to continue scheduled tumor assessments so that the date of progression can be documented.

Secondary Efficacy Outcome Measures

Secondary efficacy outcome measures for this study include the following:

- Overall survival, defined from the time of randomization

Safety Outcome Measures

Safety outcome measures for this study include the following:

- The number and attribution of all adverse events (including vital signs, physical findings, and clinical laboratory results) that occur in patients who have received CMP-135 or placebo

SAFETY PLAN

The collection of all adverse events and serious adverse events will begin on the day of randomization and will continue for 45 days after the last dose of study treatment or after the initiation of new anti-tumor therapy, whichever is earlier. Any serious adverse events that occur after informed consent but prior to randomization and that are deemed to be related to study-mandated procedures will be reported. All randomized patients will be monitored for adverse events regularly by Company staff via electronically captured data that allow for near real-time evaluation of reported events. In addition, this monitoring will include investigations of possible safety trends with respect to study drug. Serious adverse events are to be reported in an expedited manner within 24 hours to the Medical Monitor and to Company's Drug Safety Department.

Although no specific concomitant medications except for non−protocol-specified anti-tumor therapy are prohibited during this study, CMP-135 may potentially affect the pharmacokinetics of other drugs by altering their metabolism. In addition, metabolic inducers could possibly modify the pharmacokinetics of CMP-135. Therefore, concomitant medications should be used with care, and the risk-benefit profile of each agent should be taken into consideration. Appendix B lists concomitant medications that may have the potential to interact with CMP-135.

STUDY TREATMENT

Patients will receive 150 mg of CMP-135 daily by mouth or placebo daily by mouth, beginning on Day 1, continuously until radiographic disease progression, intolerable toxicity most probably attributable to CMP-135 or placebo, or withdrawal from the study.

Women of childbearing potential must have a negative pregnancy test on the day they receive their monthly supply of CMP-135 or placebo. The patient's use of two effective forms of contraception (including one barrier method) must be confirmed before dispensing CMP-135 or placebo.

CONCOMITANT THERAPY AND CLINICAL PRACTICE

Concurrent non−protocol-specified anti-tumor therapy (e.g., chemotherapy, hormonal therapy, other targeted therapy, investigational agents, radiation therapy, surgery) will not be allowed on this study. Although no specific non−anti-tumor concomitant medications are prohibited during this study, CMP-135 may potentially affect the pharmacokinetics of other drugs by altering their metabolism. In addition, metabolic inducers could possibly modify the pharmacokinetics of CMP-135. Therefore, concomitant medications should be used with care, and the risk-benefit profile of each agent should be taken into consideration. Appendix B lists concomitant medications that may have the potential to interact with CMP-135.

STATISTICAL METHODS

Primary Efficacy Analysis

The primary efficacy endpoint in this study is the hazard ratio (HR) of PFS using the stratified Cox model. PFS will be defined as the time from randomization until the first occurrence of radiographic progression or death from any cause, determined by the investigator (see Primary Efficacy Outcome Measures, Section 3.3.1, for a definition of radiographic progression). Patients who do not meet the criteria for disease progression, but who die within 45 days of last treatment dose, will be treated as having progressed for the PFS analysis. Patients who do not meet the criteria for disease progression and who do not die will be censored at the time of last tumor assessment.

The stratification factor for the stratified Cox model will be remission history (second remission/third remission). This primary analysis will include a 90% confidence interval (CI) for the stratified HR for PFS. Results from a stratified and unstratified log-rank test will also be presented. Kaplan-Meier methodology will also be used to estimate the PFS curves and medians in each treatment arm.

Missing Data

For PFS, patients who are lost to follow-up will be analyzed as censored observations on the last date that the patient was known to be progression free plus 1 day. If no post-randomization tumor assessment is available, the observation will be censored on the day after the randomization date. For duration of survival, patients who are lost to follow-up will be analyzed as censored observations on the date of last contact.

Determination of Sample Size
The sample size of approximately 219 patients for this study.

Interim Analysis
No interim analyses for efficacy are planned.

LIST OF ABBREVIATIONS AND DEFINITION OF TERMS

AE adverse event
BCC basal cell carcinoma
BUN blood urea nitrogen
CI confidence interval
CRA clinical research associate
CRO contract research organization
CT computed tomography
EC ethics committee
ECOG Eastern Cooperative Oncology Group
eCRF electronic Case Report Form
EDC electronic data capture
FDA Food and Drug Administration
GCP Good Clinical Practice
HDPE high-density polyethylene
HEENT head, eye, ear, nose, and throat
HR hazard ratio
ICH International Conference on Harmonisation
IHC immunohistochemistry
IND Investigational New Drug
IRB Institutional Review Board
ITT intent to treat
IV intravenous
IVRS interactive voice response system
IWRS interactive web response system
NCI CTCAE National Cancer Institute Common Terminology
Criteria for Adverse Events
NF National Formulary
PFS progression-free survival
PK pharmacokinetic
PS performance status
RBC red blood cell
RT-PCR reverse transcriptase polymerase chain reaction
SAE serious adverse event
SMO Smoothened (7-pass transmembrane domain protein)
ULN upper limit of normal
USP United States Pharmacopeia
WBC white blood cell

1. BACKGROUND

Ovarian cancer is a disease with ongoing, significant unmet medical need. In the United States, an estimated 21,650 new cases of ovarian cancer will be diagnosed, with an estimated 15,520 deaths in 2008 (American Cancer Society 2008). Primary peritoneal carcinoma and fallopian tube carcinoma are less commonly occurring cancers that are treated the same way as ovarian cancer. Most patients with ovarian cancer are diagnosed with advanced (Stage III or Stage IV) disease and are treated with surgery followed by either intravenous
(IV) or intraperitoneal chemotherapy. This chemotherapy typically consists of a platinum agent (cisplatin or carboplatin) combined with a taxane (e.g., paclitaxel). Although many patients respond to initial therapy, the majority will experience disease recurrence and will require additional chemotherapy. Therefore, maintenance chemotherapy has been studied in the first-line setting as a potential way to decrease or delay the recurrence of ovarian cancer after initial surgery and chemotherapy.

2. OBJECTIVES

2.1 PRIMARY OBJECTIVE

The primary objective for this study is as follows:

- To estimate the clinical benefit of the addition of CMP-135 as maintenance therapy for patients with ovarian cancer in a second or third complete remission, as measured by investigator-determined PFS by radiographic assessment

2.2 SECONDARY OBJECTIVES

Secondary objectives for this study are as follows:

- To make a preliminary assessment of the safety and tolerability of CMP-135 in this population
- To make a preliminary assessment of the efficacy of CMP-135 as measured by overall survival

2.3 EXPLORATORY OBJECTIVES

3. STUDY DESIGN

3.1 DESCRIPTION OF THE STUDY

The proposed study is a Phase II, randomized, placebo-controlled, double-blind, multicenter clinical trial of CMP-135 in patients with ovarian cancer in a second or third complete remission. Patients will be randomized in a 1:1 ratio to either CMP-135 or placebo. Randomization will be stratified based on whether their cancer is in a second or third complete remission. Patients will receive 150 mg of CMP-135 or placebo daily by mouth. Treatment will continue until evidence of radiographic progression, until the patient experiences intolerable toxicities most probably attributable to CMP-135, or until the patient withdraws from the study. Day 1 of the study will be defined as the first day the patient receives CMP-135 or placebo. Patients who discontinue study treatment (i.e., due to documented disease progression) will be followed for survival approximately every 3 months until death, loss to follow-up, or study termination by Company.

3.2 RATIONALE FOR STUDY DESIGN

This study is designed to compare CMP-135 with placebo as maintenance therapy for patients with ovarian cancer in a second or third complete remission to determine if CMP-135 can prolong PFS. The continuous dosing schedule of CMP-135 was chosen, based on the pharmacokinetic (PK) properties of CMP-135 characterized in the Phase I study of this agent, to maximize exposure and potential therapeutic effect. Along with assessments for the safety and tolerability of CMP-135 or placebo, patients will be evaluated for disease progression at predefined, standard intervals to minimize evaluation time biases.

3.3 OUTCOME MEASURES

3.3.1 <u>Primary Efficacy Outcome Measure</u>

The primary outcome measure is PFS, defined as the time from randomization to the first occurrence of radiographic progression or death from any cause, determined by the investigator.

Radiographic progression is defined as the appearance of a new lesion on a computed tomography (CT) scan of the chest/abdomen/pelvis consistent with recurrent disease, including new onset or worsening ascites. New ambiguous lesions may be biopsied to determine progression. The patient should remain on blinded study treatment (CMP-135 or placebo) until radiographic progression is confirmed. In the case of clinical deterioration, such as small bowel obstruction, a CT scan should be performed to determine whether radiographic progression has occurred. If radiographic progression cannot be documented objectively, the investigator must consult the Medical Monitor before removing a patient from the study. Increasing levels of CA-125 in the absence of radiographic progression has not been validated as an endpoint in this setting and should not be interpreted as progression or as a criterion for discontinuation of study treatment.

For patients who discontinue study treatment prior to documented radiographic progression, every effort should be made to continue scheduled tumor assessments so that the date of progression can be documented.

3.3.2 <u>Secondary Efficacy Outcome Measures</u>